

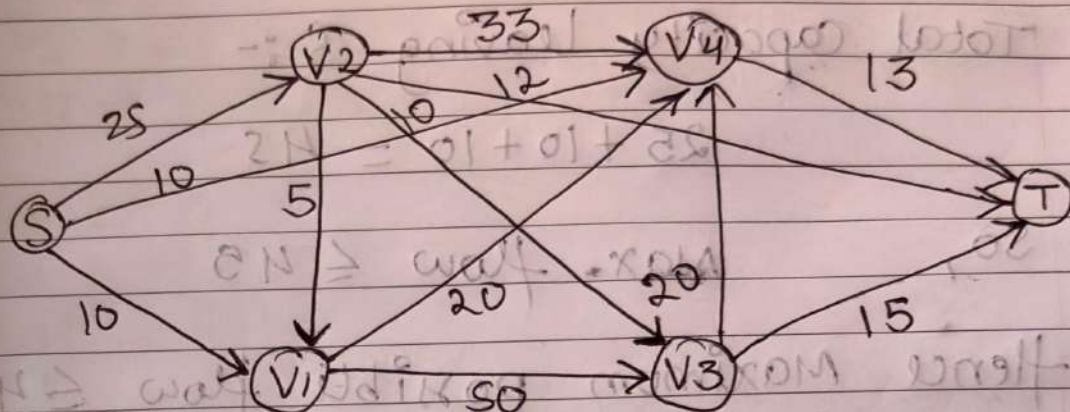
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DAA-LAB

ASSESSMENT-4

⇒ Ford Fulkerson Algorithm :-



⇒ Edges And Capacities :-

$$S \rightarrow V2 = 25$$

$$S \rightarrow V4 = 10$$

$$S \rightarrow V1 = 10$$

from V2 :-

$$V2 \rightarrow V4 = 33$$

$$V2 \rightarrow V3 = 10$$

$$V2 \rightarrow V1 = 5$$

$$V2 \rightarrow T = 12$$

from V4 :-

$$V4 \rightarrow T = 13$$

from V1 :-

$$V1 \rightarrow V3 = 50$$

$$V1 \rightarrow V4 = 20$$

from V3 :-

$$V3 \rightarrow V4 = 20$$

$$V3 \rightarrow T = 15$$

Step-1 :- Upper Bound check :- $T = 12 + 15 + 13 = 40$

• Total Capacity Entering

$$\text{Max flow} \leq 40$$

Total capacity leaving S :-

$$25 + 10 + 10 = 45$$

So, $\text{Max flow} \leq 40$

Hence Maximum possible flow ≤ 40 .

2) Step-by-step Ford Fulkerson :-

• path 1 :-

$$S \rightarrow v_2 \rightarrow T$$

$$\min(25, 12) = 12$$

flow = 12

$$\text{Remaining } S \rightarrow v_2 = 13$$

$$\text{Total} = 12$$

• path 2 :-

$$S \rightarrow v_4 \rightarrow T$$

$$\min(10, 13) = 10$$

flow = 10

$$\text{Remaining } v_4 \rightarrow T = 3$$

$$\text{Total} = 22$$

◦ path 3 :-

$$S \rightarrow V_2 \rightarrow V_4 \rightarrow T$$

Remaining :-

- $S \rightarrow V_2 = 13$
- $V_2 \rightarrow V_4 = 33$
- $V_4 \rightarrow T = 3$

$$\min = 3$$

$$\text{flow} = 3$$

$$\text{Total} = 25$$

◦ path - 4 :-

$$S \rightarrow V_2 \rightarrow V_3 \rightarrow T$$

Remaining :-

- $S \rightarrow V_2 = 10$
- $V_2 \rightarrow V_3 = 10$
- $V_3 \rightarrow T = 15$

$$\min = 10$$

$$\text{flow} = 10$$

$$\text{Total} = 35$$

◦ path - 5 :-

$$S \rightarrow V_1 \rightarrow V_3 \rightarrow T$$

Remaining :-

- $S \rightarrow V_1 = 10$
- $V_1 \rightarrow V_3 = 50$
- $V_3 \rightarrow T = 15 - 10 = 5$

$$\min = 5$$

$$\text{flow} = 5$$

$$\text{Total} = 40 \quad \left\{ \begin{array}{l} \text{final Maximum} \\ \text{flow} = 40 \end{array} \right.$$